Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 322 :: Computer Peripheral and Interfacing Sessional
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1. Introduction

There are many health and safety risks that the growing elderly population is at risk for, especially for the elderly who live alone. Falls, abnormal vital signs, and cardiac problems can be serious as they may be not detected until later consequences. The Elderly Health & Fall Detection Smart System is designed to solve these problems.To overcome these difficulties, the Elderly Health & Fall Detection Smart System is designed to address them.

It constantly maintains the clinical status of the patient with various health parameters such as the heart rate, the SpO_2 level, the body temperature and the body weight. Using integrated sensors and a Raspberry Pi, the temperature, and fall events, are measured. In emergency situations, automated alerts are sent to pre-defined contacts via calls and messages, allowing for quick response.

A web based application supports the system, which produces health reports based on the collected sensor data. The website keeps track all the patients data which helps what kind of age and weight the patient has. This information is used later when we take data from the raspberry pi or when we take the data manually.

A health parameter based machine learning model is used to give a risk assessment.Combined with an associated risk score. This allows potentially health problems to be identified at an early stage, Facilitates informed decisions for care providers and health care partners.

The overall system proposed in this paper will help improve the care of the elderly by monitoring in real time, intelligent risk assessment, and intelligent health checks.

2. Objectives

This project aims to do the following:

Real-time Health Monitoring: To monitor critical physiological parameters of the continuously. HR, SpO_2, and temperature, among other parameters, were measured in the elderly using integrated sensor modules.

Automatic Fall Detection: To detect fall incident, abnormal body motion through the use of sensors include an accelerometer and a gyroscope with a threshold-based and rule-based detection algorithms.

Emergency Alert System: To ensure timely response the alerts were automatically sent through SMS by implementing EAS. An automated dialing to preconfigured emergency contacts for abnormal and critical conditions.

Web-Based Monitoring Interface: To create an easy to use web application to visualize real time data from the health quality monitoring network. Collects patient information and computes a health risk score with the help of a trained machine learning model. This current health state and risk level to be efficiently monitored by interface.

Data Logging and Analysis: To push sensor data directly to the web from the Raspberry Pi.It is stored in a centralized database as a patient history in a real-time based system. The stored Historical data is used to conduct trend analysis, and displayed using tabular views and Graphical representations for better clinical interpretation.

GPS tracking system: Integration of GPS tracking module helps to give the exact location of problem execution via the message and telegram bot text.

Multi-channel emergency support: In critical situations the system automatically notifies all preprogrammed emergency contacts by SMS and phone and also sending the precise real-time data to a Telegram bot to respond quickly and coordinate.To ensure an emergency response system in the event of an emergency is provided, even if the primary system fails.

Secondary Contact support: The cases in which automatic alert generation does not work. In these cases, only authorized people (such as a doctor or health care worker) should be allowed to handle the medication. Stored patient data can be accessed by the caregiver (cg) and emergency contact information can be retrieved for the caregiver to manually Make an initiation call and maintain system reliability and continuity of care.

Portable and Wearable Design: To design and make a portable, low power and wearable design. A device that can provide users with a good degree of mobility and comfort while ensuring continual health monitoring.

3. Problem Statement

There are a number of important problems that the elderly have to deal with, especially those who live on their own:

Delayed Emergency Response: Fall or sudden health change is not noticed long periods with serious consequences or even death.

Limited Continuous Monitoring: The family members cannot monitor the elderly People for various reasons—distance, work, etc. So they are unable to be with others in real time.

Manually dependent: Many existing systems still rely on manual alerting intervention to ask for assistance – this can be difficult during emergency situations.

No Intelligent Risk Scoring: Most traditional monitoring solutions don't offer a unified risk assessment. For instance, a fall isn't necessarily a cause of a critical condition when other vital parameters (SpO_2, HR, temp) are normal. If a risk scoring mechanism is not used, it is challenging to assess the Patient's general physical condition.

High Cost of Professional Care: Continuous professional healthcare monitoring is too costly and unavailable to all families.This project aims to tackle these challenges by creating an affordable, automated and intelligent health monitoring system where the system detects the health condition in real time and evaluates the risk with accuracy using a machine.

No combination of ML : Most traditional system use only the sensors so we can only know the data. But when we want to predict how much emergency the situation is we fail to detect the score.The integration of machine learning in a sensor based project is very much unfamiliar at an affordable price range.

4. Scope

The project will be carried out on the following aspects:

4.1 Functional Scope

- Vital signs monitoring (heart rate, SpO_2, temperature) is continuous.
- In real time fall detection with motion sensors (accelerometer and gyroscope).
- Automated emergency alert system by SMS and by phone calls, by the use of network connectivity.
- Local warning system (Buzzer & LED) for immediate warning on site.
- Real-Time data transfer between Raspberry Pi and a web based system to which the sensor data is sent.
- stored in a centralized database in real time for historical access and analysis tracking.
- storing data in the website database to store patient history, for future retrieval and tracking.
- The generation of a risk score with a machine learning model trained with sensor data.
- Real time visualization of patient's health status and risk level via a web interface
- This allows the use of GPS to track the patient's location during normal functioning of the system and emergency conditions.That helps in automatic transmissions of emergency alerts and live location, which are sent by SMS.
- Emergency notification with accurate GPS coordinates to Telegram bot for speedy response to predefined contacts.
- The response and external monitoring support is excellent.
- Use the Raspberry Pi as a server.

4.2 Technical Scope

- Multiple sensors integration to ESP32 & Raspberry Pi that is use for communication with the device and sensors interaction.

- Develop algorithms for emergency detection that are based on thresholds and rules.
- Developed a Python backend and web interface for real-time monitoring.
- Built-in Wi-Fi for wireless setup and connection to smartphones or computers and smartphone operation.
- GPS module integration for real-time location tracking of the patient during normal operation and emergency conditions
- Live GPS transmission with emergency alerts to caregivers via SMS and phone calls in the event of an emergency, the location is automatically shared with the people who are supposed to receive it via a Telegram bot.
- RPM - scenarios for quick response and monitoring in the website with the help of Raspberry pi.
- Power management strategies to support portable and wearable operation.

4.3 Target Users

- People from an older generation who live alone or with little supervision.
- Families where senior citizens are in need of constant medical supervision.
- Health authorities and health care workers.
- Rehabilitation centers and old people's homes.
- Even people who are physically unstable or vulnerable.

5. Methodology

5.1 System Architecture

The system comprises of two major processing units operating together:

ESP32 Microcontroller:

- Data acquisition of all integrated sensors was done primarily.
- Analysis of physiological and motion data in real time.
- A set of thresholds and rules are used to identify emergencies.
- Alerts can be triggered by buzzer and LED indicators.
- Communication with Raspberry Pi through Wi-Fi protocol.
- Handling of GSM module to send SMS and to trigger call alerts.

- Use GPS module to track exact location that is given in the text.
- Real-time location tracking and make call using of 4G-SIM card functions.
- Normal and emergency patient coordinates here.

Raspberry Pi:

- To read or write data to the sensor on the ESP32 act as server.
- Running back end systems for data management and system coordination.
- Ability to communicate with the Web application in real time.
- Intermediary connections between hardware system and the online platform.
- Integrating alerts and system information to enable a client-server alarm distribution system.
- Supporting future cloud-related flexibility in data management.
- Processing data and sending to the backend APIs for risk analysis and storage.

Web-Based System Methodology:

- The web application is an integral part of the system and is designed to enhance monitoring and decision-making.
- Fetches the sensor data from the raspberry pi in real-time through its endpoints built on fast api.
- Persists patient information in a relational database (PostgreSQL) that can be accessed at all times.
- Applies trained machine learning models (XGBoost and Random Forest) to compute a health risk score that is calculated using the data from SpO_2, HR, temperature and fall detection.
- Dynamic dashboard for caregivers to watch in real-time health status and risk levels.
- Presents patient data graphically for trending analysis .Uses Chart.js for interactive and responsive visualization of data.
- Creates automatically a health survey for medical review and opinion support information in the event of an emergency.
- Provides access to emergency review, allowing authorized users to review patient information manually in case of an emergency.
- Multipurpose website that can store report history, can take data both manually and directly from raspberry pi.
- The data of patient can be accessed by central admin who can act as doctor or caregiver to monitize regularly.

Technologies Used

- **Electronic:** ESP32, Raspberry Pi, Heart Rate Sensor, SpO_2 Sensor, Temperature Sensor, Accelerometer, Gyroscope, GPS Module, GSM Module
- **Power Management:** 3.7V lithium rechargeable battery + charger, 1000mAh rechargeable battery, BQ25894, MP2550, BQ25503, BQ25504, 26W fast charge charges Li-ion batteries
- **Backend:** FastAPI (Python)
- The Frontend is HTML, CSS, JavaScript.
- **Database:** PostgreSQL
- **Machine Learning:** XGBoost, Random Forest
- **Visualization:** Chart.js

5.2 Working Methodology

Step 1: System Initialization

All the sensors and modules connected to the system are enabled when the system is turned on. The ESP32 communicates with the sensors using I2C and UART. The Raspberry Pi starts-up and starts any of the back-end services and the monitoring interface. Network connectivity test (WiFi/GSM) is conducted and ensured, to ensure data transmission and alert capacity.

Step 2: Continuous Health Monitoring

- The system continuously records the data of the old person's physiology:
- The sensor, the MAX30102, is used to measure heart rate and blood oxygen saturation (SpO_2)
- The temperature sensor will take the body temperature at regular intervals.
- All sensor readings are received and analyzed in real-time and are compared to certain safe threshold values.

Step 3: Fall Detection

The MPU6050 accelerometer and gyroscope are constantly measuring body motion and orientation. A fall event is detected when the acceleration of the device exceeds a specified value or the posture is abnormal. Several readings are taken within a short time window, to eliminate false alarms, before a fall event is finally confirmed.

Step 4 : Emergency Condition Detection

To detect the emergency condition, follow the steps below:

- Emergency is detected if any of the following conditions occurs:
- In the event of a fall, motion sensors detect it.
- Pupil behaviour is overly loud or quiet
- The blood oxygen saturation drops from normal (usually >90%).
- Thermoregulatory control is lost and the body's temperature is out of normal range.

Step 5 : Alert & Notification System

- Create an alert and notification system.Finally, establish an alert and notification system (Step 5).
- In the event of an emergency condition being confirmed:
- When the ESP32 is activated, it will sound an alarm and flash an LED for immediate local alarm.
- System automatically notifies via SMS through GSM module.
- The notifications are sent using network-based notification on Raspberry Pi.
- Alerts are also sent via Telegram message from the Raspberry Pi, if internet connection is available.

Step 6: Data Logging, Monitoring and Web Integration

All the sensor data gathered from the system is sent to the Raspberry Pi which then sends it to a FastAPI based back-end and saves it to a PostgreSQL database. This data is always updated to a central monitoring web app.

The Web system works in the following ways:

- Stores real-time and historical patient health data in a structured database collect from Raspberry .
- Graphically presents real-time health data on an interactive dashboard
- Computes health risk score using machine learning models (XGBoost and Random Forest) with the inputs of SpO_2, HR, temp and fall detection.
- Uses Chart.js graphs to visualize patient trends and patterns for improved medical interpretation
- Produces automated health reports to caregivers and healthcare professionals
- Offers historical analysis, to aid in the identification of early warning signs and long-term health trends
- Allows caregivers to track patient status and risk level remotely in real time

5.3 Software Architecture

Programming ESP32 using the Arduino C++ programming language:

- For each of the integrated modules, there are sensor data acquisition routines.
- Threshold based and rule based alert detection algorithms
- Wireless internet connection to Raspberry Pi to transfer data in real-time
- GSM module control for SMS/call alerts
- The integration of a GPS module to get real-time location in normal and emergency conditions and transmitting it.

Raspberry Pi Programming: (Python):

- Uploading files to the server and coordinating the system using python.Uploading files to the server and coordinating the system using python.
- Using FastAPI for data handling and communication with services.
- Live visualization and monitoring of data.
- Send real-time alerts and notifications to Telegram Desktop using Wi-Fi.
- Alert dissemination over the network for emergency communications.
- Managing the communication between ESP32 hardware system and web application.
- Transmitting sensor data to the web platform for analysis and risk scoring.

Web Application Architecture:

- Was developed to be a centralized health monitoring platform.
- Gets real-time sensor data from Raspberry Pi through FastAPI endpoints.
- Stores and manages patient data in a postgres database.
- Uses machine learning models (XGBoost and Random Forest) to create a health risk score.
- Enables caregivers to have a dynamic dashboard to see real-time health status and risk levels.
- Presents patient history in a visual format for trend analysis.
- Develops interactive, responsive data visualisation using Chart.js
- Creates automated health reports to support medical review and decision making
- Allows remote access to patient status, historical data and risk level.

5.4:Technology Stack

5.4.1 Hardware Technologies

Component	Description
ESP32 Microcontroller	Main edge device used for sensor data acquisition, processing, and wireless communication (Wi-Fi + Bluetooth).
Raspberry Pi	Central processing unit for data aggregation, backend communication, and system coordination.
Heart Rate Sensor (PPG Sensor)	Measures pulse rate by detecting blood volume changes in the fingertip.
SpO_2 Sensor	Measures blood oxygen saturation level (SpO_2) for patient monitoring.
Temperature Sensor (DS18B20 / DHT11)	Captures body or environmental temperature for health analysis.
Neo-6M GPS Module	Provides real-time location tracking
Accelerometer / Fall Detection Sensor	Detects sudden movements or falls for emergency alert generation.
GSM Module	Provides SMS/call-based emergency alerts when internet is unavailable.
Power Supply Unit	Provides stable power to ESP32 and Raspberry Pi systems.

5.4.2 Software Technologies

5.4.2.1 Rasberry Pi Technologies

Technology	Purpose
Python	Core programming language for backend processing and system control.
FastAPI	Lightweight backend framework for receiving and sending sensor data via APIs.
PostgreSQL	Database system used for storing real-time and historical patient data.
Telegram Bot API	Sends real-time alerts and notifications to caregivers via Telegram.
Pandas / NumPy	Used for handling, cleaning, and processing sensor data.

5.4.2.2 ESP32 Technologies

Technology	Purpose
Arduino C++	Programming language used for ESP32 firmware development.
Wi-Fi (ESP32 Built-in)	Enables wireless communication with Raspberry Pi backend.
Sensor Libraries	Interfaces for reading heart rate, SpO_2, temperature, and motion
MQTT / HTTP Protocol	Used for sending sensor data to Raspberry Pi in real time.
Interrupt-Based Programming	Ensures fast response for emergency events like fall detection or threshold alerts.

5.4.2.3 Web Application Technologies

Technology	Purpose
HTML, CSS, JavaScript	Frontend structure, styling, and interactivity of the dashboard.
Chart.js	Visualizes patient health trends and real-time
FastAPI Integration	Connects frontend with backend data services.
PostgreSQL (via API)	Fetches and displays stored patient records.
Machine Learning Models (XGBoost, Random Forest)	Generates health risk scores based on sensor inputs.

5.4.3 Communication Technologies

The I2C is a short-range protocol that is used to communicate between sensors and ESP32s.

- **UART Protocol:** Serial communication for GSM ,GPS Module and debugging.

- **Wi-Fi Network:** Main communication between ESP32 and Raspberry Pi for real time data transfer.
- **GSM Network:** Back-up communication for SMS/call alerts in an emergency or when the network is down.

6.Hardware Components

Category	**Component name**	**Unit Price (BDT)**	Qty	Total (BDT)
Controllers	Raspberry Pi 4 Model B 4GB	14000	1	15000
	Arduino Uno R3	1000	1	1000
	ESP32 NODEMCU Development board	630	1	630
Sensors	ADXL335 Module 3-axis Analog Output Accelerometer GY-61	630	1	630
	Pulse Sensor 3.3-5V Heartbeat Heart Rate Monitor Sensor	550	1	550
	LM35D Analog Temperature Sensor Module 5V	150	1	150
	SIM800L Mini GPRS GSM Module	400	1	400
	MAX30102 Heart Rate and Pulse Oximeter Sensor Module	350	1	350

	U-Blox NEO-6M GPS Module	450	1	450
Power& Audio	Breadboard Power Supply Module 3.3V 5V	80	1	80
	MCP3008 - Analogue to Digital Converter	570	1	570
	3.7V 18650 Rechargeable Battery Solderable	95	2	190
	18650 Battery Holder – 1 Cell	50	2	100
	Official Power Supply for Raspberry Pi 4 (White)	1450	1	1450
	50 cm Cable For Arduino UNO/MEGA (USB A to B)	130	1	130
	Micro HDMI Male to HDMI Male Cable for Pi 4 or 5	385	1	385
	PCB Mount Mini Speaker - 8 Ohm 0.2W	120	1	120
Mechanical & Misc	Breadboard Full Size	150	2	300

	Breadboard Half Size	90	2	180
	1.3 Inch I2C OLED Display Module 4pin-White	630	1	630
	Ohm (Ω) 1/4w Resistors - Pack of 5	5	4	20
	5 mm Red LED (Pack of 5)	5	1	5
	Push Button 4 pin	5	1	5
	Hardware	1000		1000
	Jumper Wire 40 Pcs Set 20cm	100	2	200
Grand Total Estimated Cost				23525tk

6.1 Project Financial Summery

Category	Cost (BDT)
Total Components Cost	23,525
Miscellaneous & Contingency	900
Grand Total Project Cost	BDT 24,425

6.2 Cost Analysis

An estimated total project cost of the proposed Elderly Health Monitoring System is BDT 24,425. The primary budget goes into the following core computing units: Raspberry Pi 4 Model B, ESP32 and Arduino UNO, that enable real-time processing and communication as well as system coordination. Many of the expenses are invested in sensor modules, such as the heart rate monitor, SpO_2, temperature, accelerometer and GPS/GSM sensor, which are crucial for precise health monitoring and emergency detection. System stability, portability and long-term operation are provided by power management components, mechanical materials and miscellaneous items. Overall, budget is optimized to balance performance and affordability, provide for successful implementation of critical capabilities, including real time monitoring, alert systems, and location tracking. The schema diagram shows the structure of the database used in the web application and the relationships between the various tables. It illustrates the logical relationships of patient data, sensor data, alerts and user data in the PostgreSQL database. Primary keys, foreign keys and relationships are shown in the diagram for consistency and integrity of the data. It also provides a visual representation of the flow of data from one entity to another, which can be used for real-time monitoring and analysis.

6.3 Cost Comparison With Other Related Projects

Feature / System Aspect	Proposed System	Typical Low-Cost Projects	Commercial/Advanced Systems
Total Cost	**BDT 24,425**	BDT 15,000 – 25,000	BDT 30,000 – 70,000+
Heart Rate & SpO_2 Monitoring	Yes (MAX30102 + Pulse Sensor)	Basic heart rate only	Advanced medical-grade sensors
Temperature Monitoring	Yes	Yes	Yes
Fall Detection	Yes (Accelerometer + Gyroscope)	Limited or basic only	Advanced multi-sensor systems
GPS Tracking	Yes	Not available in most cases	Yes
GSM Emergency Alerts	Yes (SMS + Call)	SMS only or not available	Full IoT alert systems
Machine Learning Risk Analysis	Yes (XGBoost, Random Forest)	Not available	Advanced AI-based systems
Web Dashboard	Yes (Real-time + History)	Basic dashboard or none	Fully integrated platforms
Data Storage	PostgreSQL (local/web)	Limited/local storage	Cloud-based scalable storage
Telegram Alert System	Yes	Not available	Sometimes available
System Complexity	Medium	Low	High

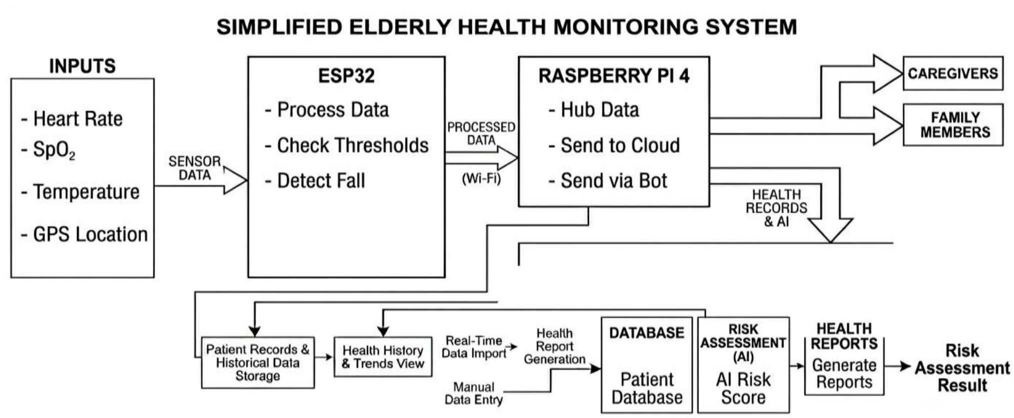
7. Work Plan (Timeline)

The development is divided into phases to ensure proper integration of hardware and software systems.

Task	Month 1	Month 2	Month 3	Month 4
Component Procurement & Testing	✓			
Circuit Design & Breadboard Assembly	✓	✓		
ESP32 Sensor Integra>on		✓	✓	
Demonstrate the capability to develop algorithms to detect falls		✓	✓	
Programming an interface for Raspberry Pi. Raspberry Pi GUI Programming			✓	✓
GPS & GSM Module Integra>on			✓	✓
Alert System Implement>on			✓	✓
System Integration & Testing				✓
Enclosure Design & Final Assembly				✓
Documenta>on & Presenta>on				✓

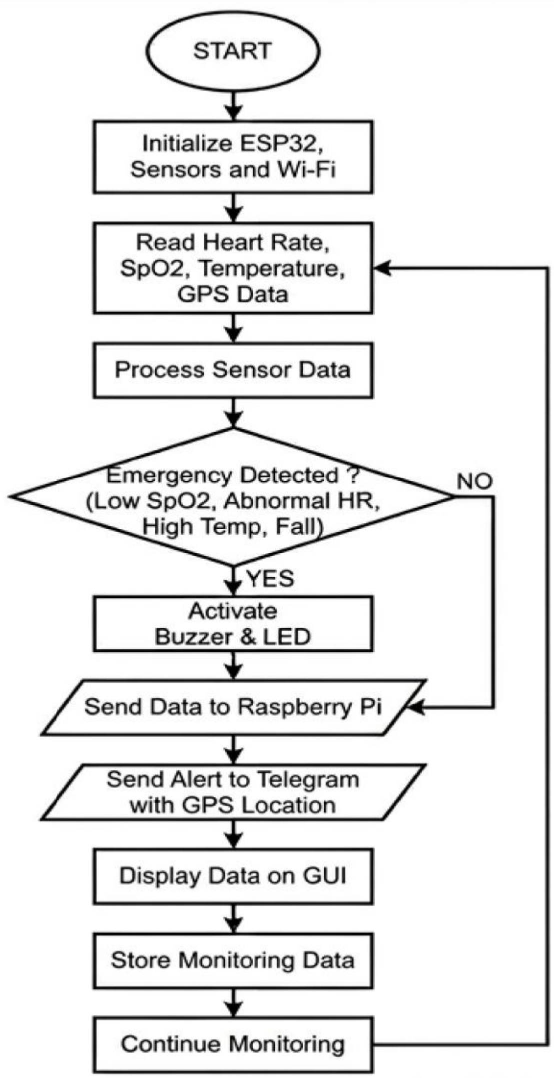
8.Visual Models

8.1.Block Diagram



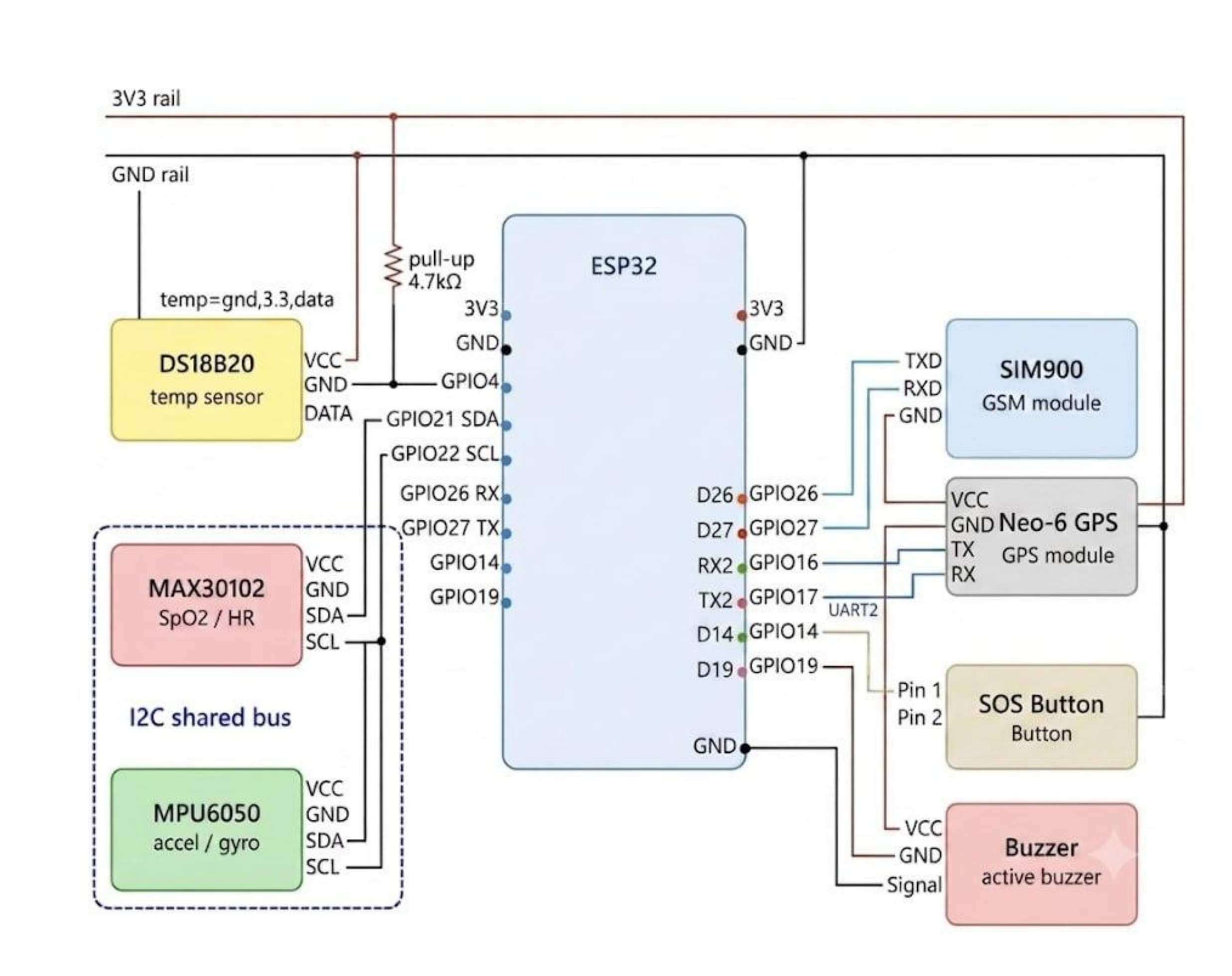
The overall system architecture and the connection of the various components are described in the block diagram. It shows the interaction network of sensors, ESP32, Raspberry Pi, GPS module and communication systems. It also demonstrates the flow of data within the hardware units and to the cloud platforms and users. It gives an overview of the overall design of the system.

8.2. Flow Chart



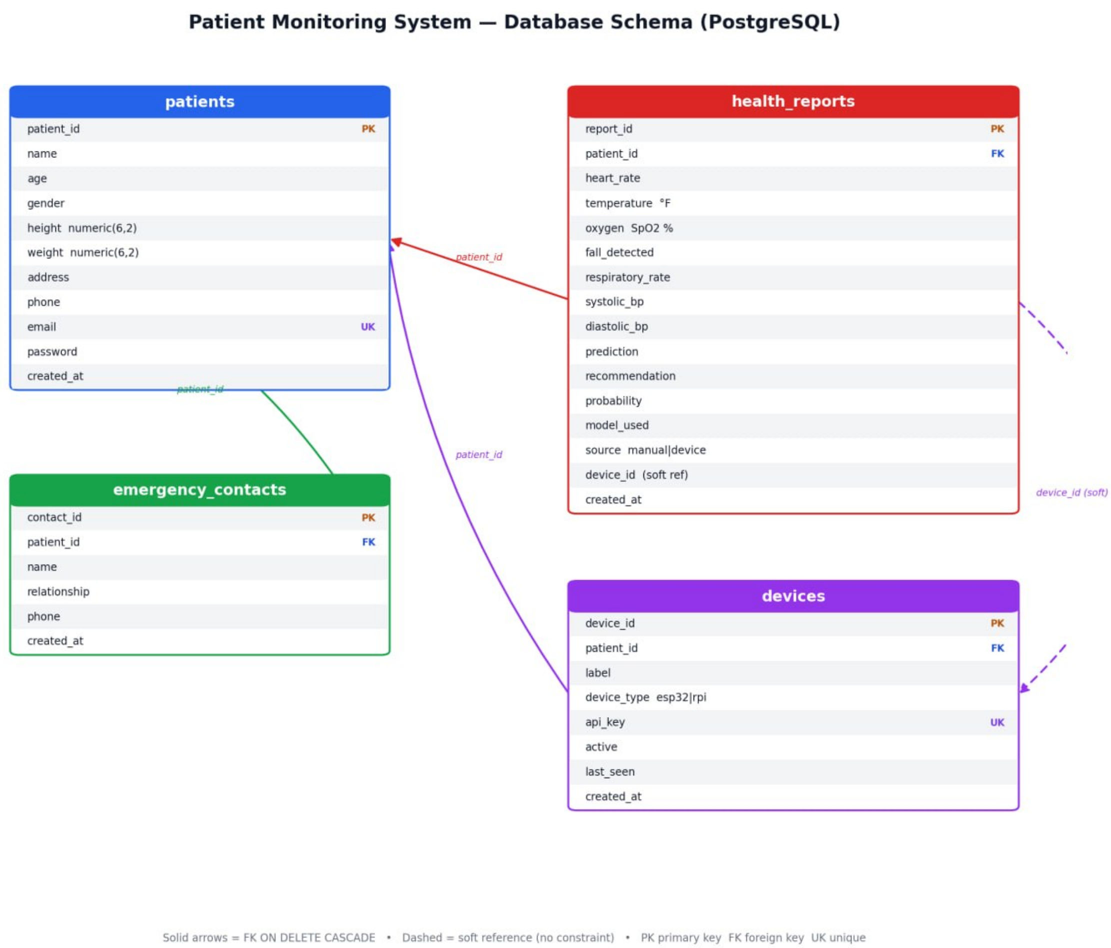
The flowchart shows the process of the system starting up and taking steps to monitor it on a regular basis. It illustrates the process of data collection, processing and comparing the sensor data against the emergency conditions. It also focuses on the pathway to make a decision from normal operation to emergency response. As a whole, it describes the logic of the control flow of the system.

8.3 Circuit Diagram



The circuit diagram gives a detailed wiring diagram of all the hardware connections. It Contains explanations on how to interface sensors such as MAX30102, LM35 and GPS with the ESP32 via I2C, and analog, and UART connections. Power supply and communication wiring between modules. It assists to understand the physical implementation of the system.

8.4 Schema Diagram



The schema diagram is the structure of the database being used in the web application: shows how tables are related to each other. It provides an illustration of how patient information and sensor data can be used to: The alerts and user information are logically related in the PostgreSQL database. The diagram Explain primary key, foreign key and the relationships to assure consistency and integrity of data. It also helps visualize how data flows between different entities for real-time monitoring and analysis.

9. Expected Outcomes

If this project is successful, the outcomes of the project are:

- A fully functional wearable health monitoring system that can track vital signs in real-time.
- Minimal false alarms, accurate fall detection
- Automated emergency alert system equipped with GPS location sharing.
- Easy to use monitoring interface for carers and health care professionals

The web-based monitoring platform should have the following features and be fully functional:

- Real-time patient health dashboard
- Health risk score generation based on sensor data with AI technology.
- Customized health information and suggestions for the patient's condition
- Historical data monitoring and trend analysis for long monitoring
- Automated health report generation for medical review.
- Automated health report generation for caregiver.

Thorough documentation, such as circuit diagrams, source code, and manuals.

Achievement of integration of several sensors, processing units, and cloud/web systems

Affordable and scalable technology solutions for elderly care and smart health solutions has been demonstrated.

10.Applications of the Project

Finally, the applications of the Project will be discussed.

- Facilitates the monitoring of elderly health in real-time, including vital signs like HR, SpO_2, temperature etc.
- Successfully integrates multiple sensors with ESP32 to enhance IoT based healthcare systems.
- For greater accuracy and reliability, use Raspberry Pi.
- Introduces a dual emergency alert system (GSM and Telegram) for quicker and more reliable communication.
- Incorporates GPS location awareness into complementing the traditional alert-only systems for emergency response.
    - Incorporates machine learning models (XGBoost & Random Forest) for health risk prediction.
    - Offers an online monitoring system to access real-time and historical data by caregivers.
    - Allows for automated fall detection, with a minimum demand of manual supervision.
- Provides an affordable replacement to the commercial health monitoring systems having advanced features.
    - Supports data logging and visualization to analyze trends and track patient health.
    - Connects the existing healthcare systems based on basic IoTs with smart monitoring solutions powered by AI for elderly care.

11. Future Plans

These improvements are planned for future releases:

- **Mobile Application Development**: Create customized Android and iOS apps for greater caregiver access
- **Voice Assistance:** Add voice control and sound feedback for helping older users use this device.
- **Multiple User Support:** Allow multiple patients to be monitored from a single screen.
- **Advanced Sensors:** Include ECG monitoring and blood pressure measurement modules.
- **Battery Optimization:** Enhance power management for longer battery life
- **Miniaturization:** Make the device smaller and easier to wear and to be comfortable in Direct Doctor to Doctor Video Consultation in Emergency situations.

12. Conclusion

The Elderly Health & Fall Detection Smart System is a crucial development in the field of IoT applications in medical care and security. This project addresses several key problems in the current elderly monitoring solutions by integrating several health sensors, intelligent processing, and automatic alerting.

It offers continuous, non-obtrusive monitoring of health and allows for instant response in case of an emergency with automated location alerts. The modular design provides the flexibility for future upgrades and scalability, catering to different healthcare settings.

Once implemented, this solution can be used as a blueprint for everyday affordable and accessible healthcare technology that can be adopted in the home, care centres and community, ultimately to enhance the quality of life and potentially save lives.

13. Related Papers

1.Fall Detection and Pulse Monitoring Alert System for Healthcare Applications using IoT Technologies https://irjaeh.com/index.php/journal/article/view/1512

Offers a solution that monitors and detects a fall using ESP32 and notifies instantly.

2.A Fall Detection and Caregiver Notification System based on the ESP32 for IoT Application https://doi.org/10.47772/IJRISS.2025.909000758

Wearable ESP32 system and emergency notification with motion sensors.

3.An IoT Based Device-Type Invariant Fall Detection System – https://www.sciencedirect.com/science/article/pii/S2542660519301623 – Accelerometer data, centralized processing for fall detection.

5.Smart Shield: IoT Based Fall Detection, GPS Tracking, and Health Monitoring System using ESP8266 https://www.ijariit.com/manuscript/smart-shield-iot-based-fall-detection-gps-tracking-and-health-monitoring-system-using-esp8266/

Implements GPS + GSM + health monitoring for emergency response.

Multi-Parameter Health Monitoring System Using IoT https://www.ijraset.com/research-paper/multi-parameter-health-monitoring-system Raspberry Pi based system for monitoring multiple parameters which sends alerts.